



Nguyen Ngoc Hoang Duong

Software Developer

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SUMMARY

Careful, responsible, and detail-oriented Software Engineering student. I prefer to understand a problem clearly before implementing it, keep my work organised, and verify important behaviour instead of assuming that it works. I am willing to learn unfamiliar technologies when a project requires them, while still paying attention to readable code, consistent interfaces, and complete results.

TECHNICAL SKILLS

Frontend

Vue 3, Vite, React, Flutter, Dart, TypeScript, JavaScript and responsive interface development

Backend

Node.js, Express, REST APIs, Socket.IO, JWT, authentication and asynchronous processing

Applied AI and Data

Python, custom RAG pipelines, nomic-embed-text embeddings, prompt construction, structured-output validation, Docling, OCR, evidence validation, evaluation and Ollama

Databases

MongoDB, PostgreSQL, MySQL, Firebase, Room DB and SQLite

Tools and Deployment

Git, GitHub, Docker, Postman, Render, MongoDB Atlas, Cloudinary and Firebase Storage

EDUCATION

2023 - 2026

University of Greenwich Vietnam

Computing - Software Engineering

- Current GPA: 3.5 / 4.0 (before the final project grade is included)
- Studied Data Structures and Algorithms, Software Engineering, Agile Development with Scrum, Advanced Microservices Development, and Data and Web Analytics.
- Completed and successfully defended DreamScape as the final-year project.

WORK EXPERIENCE

Sep 2025 - Dec 2025

Techtack

Frontend Developer Intern

- Joined the ESA Pay project during its early development and implemented Vue 3 interfaces for accounting and banking workflows.
- Worked with backend and testing tasks under frequent deadlines, while reporting progress and integration issues each day.

SELECTED PROJECTS

Sep 2025 - Dec 2025

ESA Pay

Frontend Developer Intern at Techtack

- **Business workflows:** implemented receivables, payables, accounting vouchers, invoices, and bank-related flows after studying product and API documentation.
- **Large API integration:** connected more than 100 REST endpoints to Vue screens and handled data mapping, validation, loading, empty, and error states.
- **Banking tests and data alignment:** tested banking APIs, compared payloads with provider documents, identified mismatches between frontend and backend structures, and proposed field or schema adjustments.
- **Delivery discipline:** met frequent deadlines, reported progress daily, and raised integration blockers early during the project's initial development.

The source code and project materials remain private company property.

Feb 2026 - Apr 2026

Expense Tracker System

Mobile and Full-stack Developer

- **Offline-first user application:** used Flutter and Dart for one mobile codebase, while Room DB and SQLite kept expense entry and history available without an internet connection.
- **Native administration application:** used Android and Java to implement account management, monitoring, and financial analytics while practising platform-specific application design.
- **Realtime synchronisation:** connected Firebase Authentication, Firestore, and Realtime Database through a custom queue so local changes were preserved offline and synchronised when connectivity returned.
- **HCI decisions:** kept transaction actions predictable, used clear form validation and immediate feedback, and separated user and administrator workflows so each interface displayed only the controls needed for its role.



User app repository



Admin app repository

Apr 2026 - Aug 2026

DreamScape

 Live product

Project Owner, Full-stack Developer and AI/RAG Pipeline Developer

- **Responsive and HCI design:** built one EN-VI Vue 3 and TypeScript interface for desktop and mobile, with consistent navigation, visible processing states, confirmations, and citations placed near AI-supported content.
- **Modular backend:** separated Express and Mongoose code into authentication, dream, community, messaging, academic knowledge, and Oracle modules without the operational cost of microservices.
- **Automated source pipeline:** resolved DOI metadata and HTML or XML full text, then used PDF, Docling, and OCR fallbacks. Cleaned sections were chunked and embedded before administrator review.
- **Rule and evidence automation:** extracted candidate rules after approval, linked exact evidence passages, matched new sources to unresolved evidence needs, and kept unsupported claims as visible gaps.
- **Grounded AI analysis:** used nomic-embed-text and custom RAG retrieval to select reviewed rules, evidence, personal history, and permitted similar dreams before Oracle generation; backend checks validated structure and citations.
- **Realtime infrastructure and deployment:** combined Socket.IO, MongoDB Atlas, Redis, Cloudinary, and Firebase Storage; deployed the Vue site and Express API on Render, with Ollama, embeddings, Docling, and OCR on a remote NVIDIA T4 environment.

 Frontend repository  Backend repository

ACTIVITIES

May 2025 - Apr 2026

Greenwich IT Club

Head of External Relations

Built relationships with companies and universities, managed outreach, and co-organised technology events for the student community.

CERTIFICATIONS

2022

IELTS 5.5

HONORS AND AWARDS

Top 3 students of trimester

Spring 2025 and Summer 2025

Best student by subject

Mathematics for Computer, Object-Oriented Programming, User Interface Design, and Application Development

INTERESTS

- **Sports:** badminton, gym training, and swimming.
- **Technology and hardware:** custom PC building, hardware optimisation, audio equipment, and wearable technology.
- **Design and craftsmanship:** structural design, material quality, and practical product aesthetics.